CISplatin (50mg/m^2) and Etoposide (50mg/m^2) and Thoracic Radiotherapy (TRT) -28 day

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Stage III non-small cell lung cancer (NSCLC)	C34	00456a	N/A

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patient's individual clinical circumstances.

Induction: CISplatin is administered on day 1 and day 8 and etoposide is administered on five consecutive days (Days 1-5) of a 28 day cycle for 2 cycles concurrently with radiotherapy (45Gy) unless disease progression or unacceptable toxicity develops whichever is first.

In the absence of progressive disease, two additional cycles of CISplatin and etoposide with concurrent radiotherapy may be administered.

Radiotherapy usually starts within 24 hours of the first day of chemotherapy.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

Admin Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1-5	Etoposide[i]	50mg/m^2	IV Infusion	1000mL 0.9% NaCl over 1 hour	Repeat every 28 days
2	1, 8	CISplatin	50mg/m^2	IV Infusion	1000mL 0.9% NaCl over 1 hour (pre and post hydration therapy required)[a]	Repeat every 28 days

[a] **Pre and post hydration therapy required for CISplatin**
See local hospital policy recommendations.
Suggested prehydration for CISplatin therapy:
The administration of etoposide in 1000mL 0.9% NaCl over 1 hour as detailed above may be considered as pre-hydration for CISplatin

- Administer CISplatin as described above

Post hydration:

- Administer 10mmol magnesium sulphate ($MgSO_4$) and 20mmol potassium chloride (KCl) in 1000mL 0.9% NaCl over 2 hours (Refer to relevant local hospital policy for advice on administration of electrolyte infusions).

Mannitol 10% may be used as per local policy to induce diuresis, although there is no conclusive evidence that this is required. The routine use of furosemide to increase urine flow is not recommended unless there is evidence of fluid overload.

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

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ELIGIBILITY:

- Indications as above
- ECOG status 0-1
- Suitable candidate for thoracic radiation

CAUTIONS:

- Severe liver impairment (etoposide)

EXCLUSIONS:

- Hypersensitivity to etoposide, CISplatin or any of the excipients
- CISplatin
  - Pre-existing neuropathies ≥ grade 2
  - Pre-existing renal impairment
  - Significant hearing impairment/tinnitus
- Pregnancy
- Breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Audiology and creatinine clearance if clinically indicated

Regular tests:

- FBC Day 1 and Day 8
- Renal and liver profile prior to each cycle

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant

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Haematological:

Table 1: Dose modification of etoposide for haematological toxicity

ANC (x 10^9/L)		Platelets (x 10^9/L)	Dose Etoposide
≥ 1.5	**and**	≥ 100	100%
1-1.49	**or**	75-99	75%
< 1	**or**	< 75	DELAY

Renal and Hepatic Impairment:

Table 2: Dose modification in renal and hepatic impairment

Drug	Renal impairment		Hepatic Impairment	
Etoposide	**CrCl (mL/min)**	**Dose**	**Bilirubin (micromol/L)**	**Dose**
	>50	No need for dose adjustment is expected.	Bilirubin < 50 and normal albumin and normal renal function	No need for dose adjustment is expected
	10-50	75% of the original dose, increase if tolerated.	Bilirubin ≥50 or decreased albumin levels.	Consider 50% of the dose, increase if tolerated.
	Haemodialysis	Not dialysed, consider 75% of the original dose.		
CISplatin	**CrCl (mL/min)**	**Dose**	No need for dose adjustment is expected	
	50-59	75% of the original dose		
	40-49	50% of the original dose		
	<40	Not recommended		
	Haemodialysis	50% of the original dose may be considered		

Etoposide: Renal and hepatic from Giraud et al.
CISplatin: Renal (curative intent) and hepatic from Giraud et al 2023.

Non-Haematological Toxicity:

Table 3: Dose modification schedule based on adverse events

Adverse reactions	Recommended dose modification
Grade ≥ 2 peripheral neuropathy	Substitute CARBOplatin AUC 5 or 50% reduction of CISplatin dose after recovery to grade ≤ 1; 100% dose of etoposide.
Grade 3 (Other than mucositis or alopecia)	Delay until recovery to Grade 1. Then reduce dose of CISplatin and etoposide to 75%.

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting - Available on the NCCP website

CISplatin: High (**Refer to local policy)**
Etoposide: Low (**Refer to local policy)**

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For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists. Information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for Inclusion in NCIS (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS:

Hydration pre and post CISplatin administration **(Reference local policy or see recommendations above).**

OTHER SUPPORTIVE CARE: No specific recommendations

ADVERSE EFFECTS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

1. Albain K et al. Concurrent CISplatin, Etoposide, and Chest Radiotherapy in Pathologic Stage IIIB Non–Small-Cell Lung Cancer: A Southwest Oncology Group Phase II Study, SWOG 9019. J Clin Oncol 2002; 20: 3454-3460
2. Irish Medication Safety Network: Best Practice Guidelines For the Safe Use of Intravenous Potassium in Irish Hospitals Available at: https://imsn.ie/wp-content/uploads/2020/10/IMSN-Best-Practice-Guideline-on-IV-Potassium-Oct-2020-approved.pdf
3. Prevention and management of CISplatin induced nephrotoxicity eviQ ID: 184 v.4. Available at:https://www.eviq.org.au/clinical-resources/side-effect-and-toxicity-management/prophylaxis-and-treatment/184-prevention-and-management-of-cisplatin-induced
4. Perazella MA et al. CISplatin nephrotoxicity. UptoDate. Last updated June 2024. Available at: https://www.uptodate.com/contents/cisplatin-nephrotoxicity?search=portilla%20cisplatin&source=search_result&selectedTitle=4%7E150&usage_type=default&display_rank=4
5. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
6. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at:https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
7. CISplatin 1mg/mL Concentrate for Solution for Infusion. Summary of Product Characteristics Last updated: October 2024. Accessed 27/02/2025. Available at https://www.medicines.ie/medicines/cisplatin-1-mg-ml-concentrate-for-solution-for-infusion-latex-free-vial-stopper--35271/spc

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8. Etoposide 20 mg/mL Concentrate for Solution for Infusion Summary of Product Characteristics. Last updated: 13/03/2024. Accessed 27/02/2025. Available at https://assets.hpra.ie/products/Human/28093/1b01ff56-a902-407a-bc7a-5f3452480d04.pdf

Version	Date	Amendment	Approved By
1	18/12/2017		Prof Maccon Keane
2	08/01/2020	Reviewed. Standardised treatment table	Prof Maccon Keane
3	24/06/2021	Updated hydration protocol for CISplatin	Prof Maccon Keane
4	13/05/2025	Regimen reviewed. Updated treatment section. Added a caution section and updated exclusion criteria. Updated dose recommendations for renal and hepatic impairment in align with Giraud et al. Removed adverse effects/regimen specific complications and drug interaction as per NCCP standardisation.	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

[i:] This is an unlicensed indication for the use of etoposide in Ireland. Patients should be informed of this and consented to treatment in line with the hospital's policy on the use of unlicensed medication and unlicensed or "off label" indications. Prescribers should be fully aware of their responsibility in communicating any relevant information to the patient and also ensuring that the unlicensed or "off label" indication has been acknowledged by the hospital's Drugs and Therapeutics Committee, or equivalent, in line with hospital policy.

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